

Probability Distributions

A Complete, Exam-Ready Study Guide

PMF · PDF · CDF · Bernoulli · Binomial · Poisson · Normal · Uniform · Log-Normal · Power Law · Pareto · CLT · Estimation

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1. Probability Distribution Functions

Definition: A probability distribution function describes **how probabilities are spread over the possible values** of a random variable. Think of it as a "map" that tells you how likely each outcome (or range of outcomes) is.

There are **three** main tools used to describe distributions:

Function	Used For	What it gives
PMF — Probability Mass Function	Discrete random variables	Probability of an <i>exact</i> value: $\Pr(X = x)$
PDF — Probability Density Function	Continuous random variables	A density curve; probability = <i>area under the curve</i> over a range
CDF — Cumulative Distribution Function	Both discrete & continuous	Probability that X is <i>less than or equal to</i> a value: $\Pr(X \leq x)$

1.1 Probability Mass Function (PMF) — Discrete Random Variable

Example: Rolling a fair die, $X = \{1, 2, 3, 4, 5, 6\}$. Every face is equally likely:

$$\Pr(1) = \Pr(2) = \Pr(3) = \Pr(4) = \Pr(5) = \Pr(6) = 1/6$$

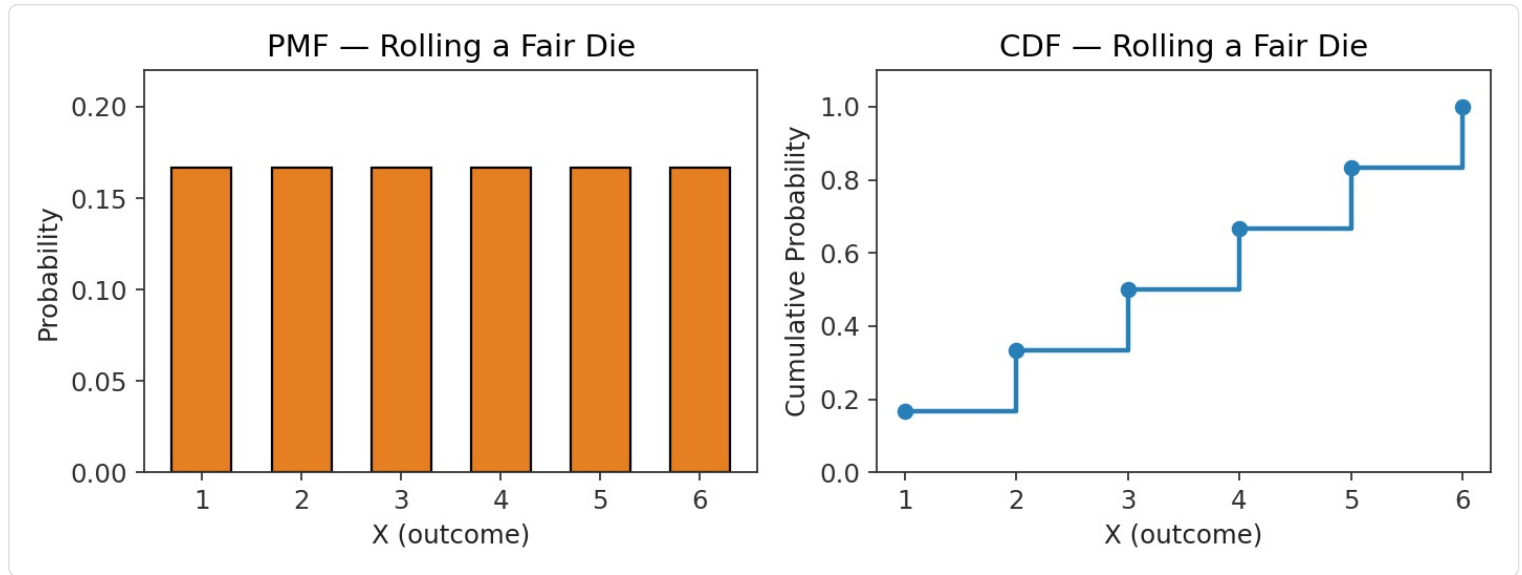


Fig 1.1 — PMF (left) shows probability of each exact outcome. CDF (right) accumulates probability as X increases.

Worked Example

$\Pr(X \leq 2) = \Pr(X=1) + \Pr(X=2) = 1/6 + 1/6 = 2/6 = 1/3$

$\Pr(X \leq 6) = \text{sum of all probabilities} = 6/6 = 1$ (certainty)

1.2 Probability Density Function (PDF) — Continuous Random Variable

For continuous variables (like age, height, weight) we cannot ask "what is $\Pr(X = 40)$?" because there are infinitely many possible values — that probability is technically 0. Instead we ask for the probability that X falls in a **range**, which equals the **area under the PDF curve** in that range.

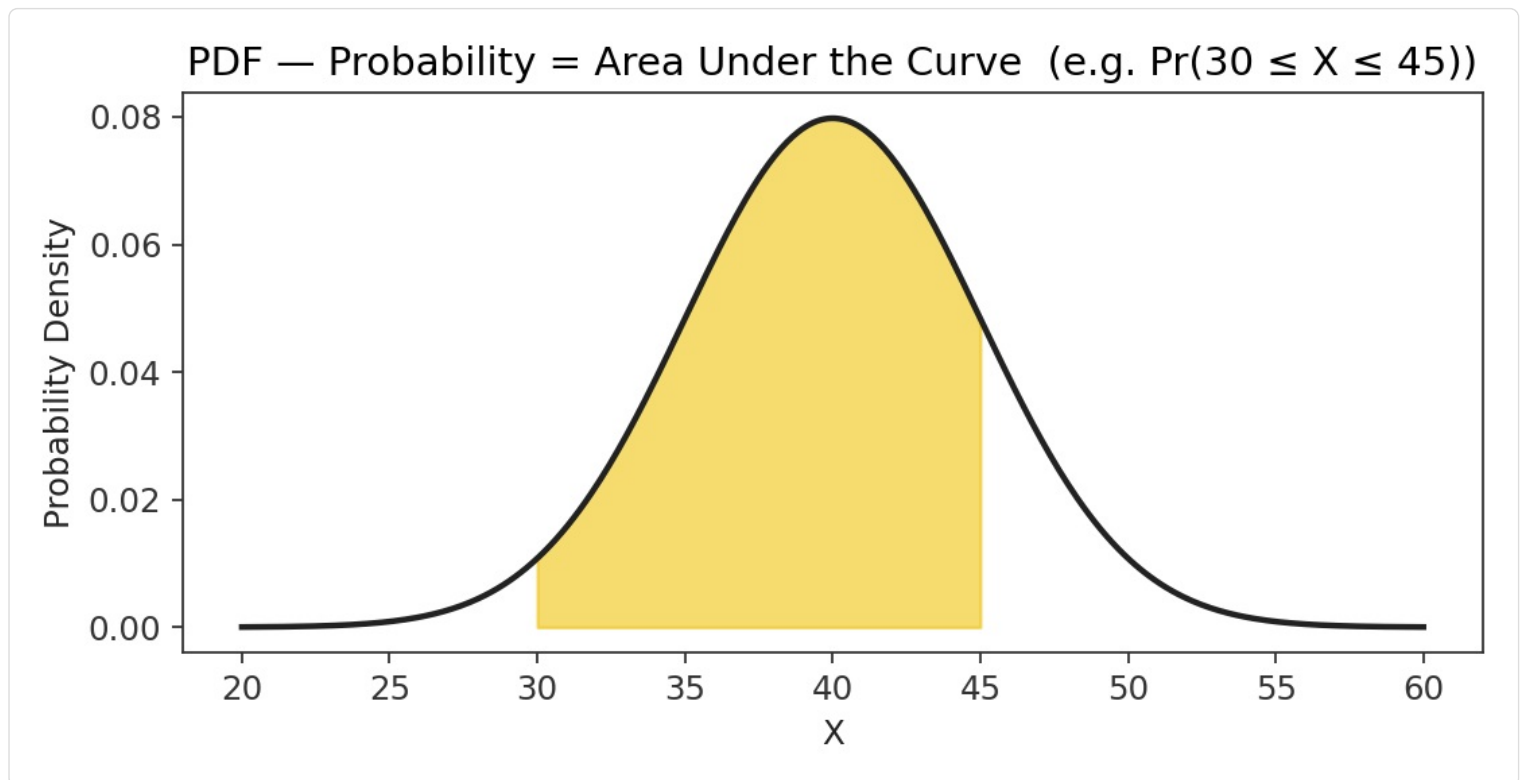


Fig 1.2 — The shaded area between 30 and 45 IS the probability $\Pr(30 \leq X \leq 45)$.

Key relationship: Probability Density = Gradient (slope) of the Cumulative Density Function.

In calculus terms: $f(x) = dF(x)/dx$, and conversely $F(x) = \int f(x) dx$

Properties of a valid PDF

1. **Non-negativity:** $f(x) \geq 0$ for all x
2. **Total area = 1:**

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

This makes sense — the probability that X takes *some* value at all must be 100%.

💡 **Tip:** The shape of $f(x)$ changes completely depending on which distribution you're dealing with (Normal, Log-Normal, Uniform, etc.) — but the "area = 1" rule never changes.

1.3 Choosing the Right Distribution Type

Variable Type	Function Used	Examples
Discrete, binary outcome	PMF (Bernoulli)	Pass/Fail, Yes/No, Heads/Tails
Discrete, count of successes	PMF (Binomial)	Number of defective items, number of correct answers
Discrete, event counts over time	PMF (Poisson)	Customers arriving per hour, website hits per minute
Continuous, symmetric	PDF (Normal)	Height, weight, exam scores, measurement error
Continuous, right-skewed	PDF (Log-Normal / Power Law)	Income, stock prices, city population

✳ Extra Example (Real Dataset Framing)

Suppose you're building a **House Price Prediction** dataset:

- **Size of house** (sq. ft.) → Continuous → PDF
- **Number of rooms** → Discrete → PMF
- **Sea-facing (0/1)** → Discrete, binary → PMF (Bernoulli-type)
- **Price** → Continuous → PDF

This is exactly the kind of reasoning a data scientist uses during EDA (Exploratory Data Analysis) and Feature Engineering — deciding whether a column is discrete or continuous decides which statistical tools apply to it.

2. Bernoulli Distribution

Definition: The simplest discrete probability distribution. It models a random variable with exactly **two** possible outcomes — success (probability p) and failure (probability $q = 1 - p$). Used for binary outcomes like a coin flip or yes/no question.

Property	Value
Type	Discrete Random Variable (PMF)
Outcomes	Binary $\rightarrow k \in \{0, 1\}$
Parameter	$0 \leq p \leq 1$ (probability of success), $q = 1 - p$

PMF: $\Pr(K = k) = p^k (1 - p)^{1-k}$ for $k \in \{0, 1\}$

Simplified: $\Pr(K=1) = p$ and $\Pr(K=0) = q = 1 - p$

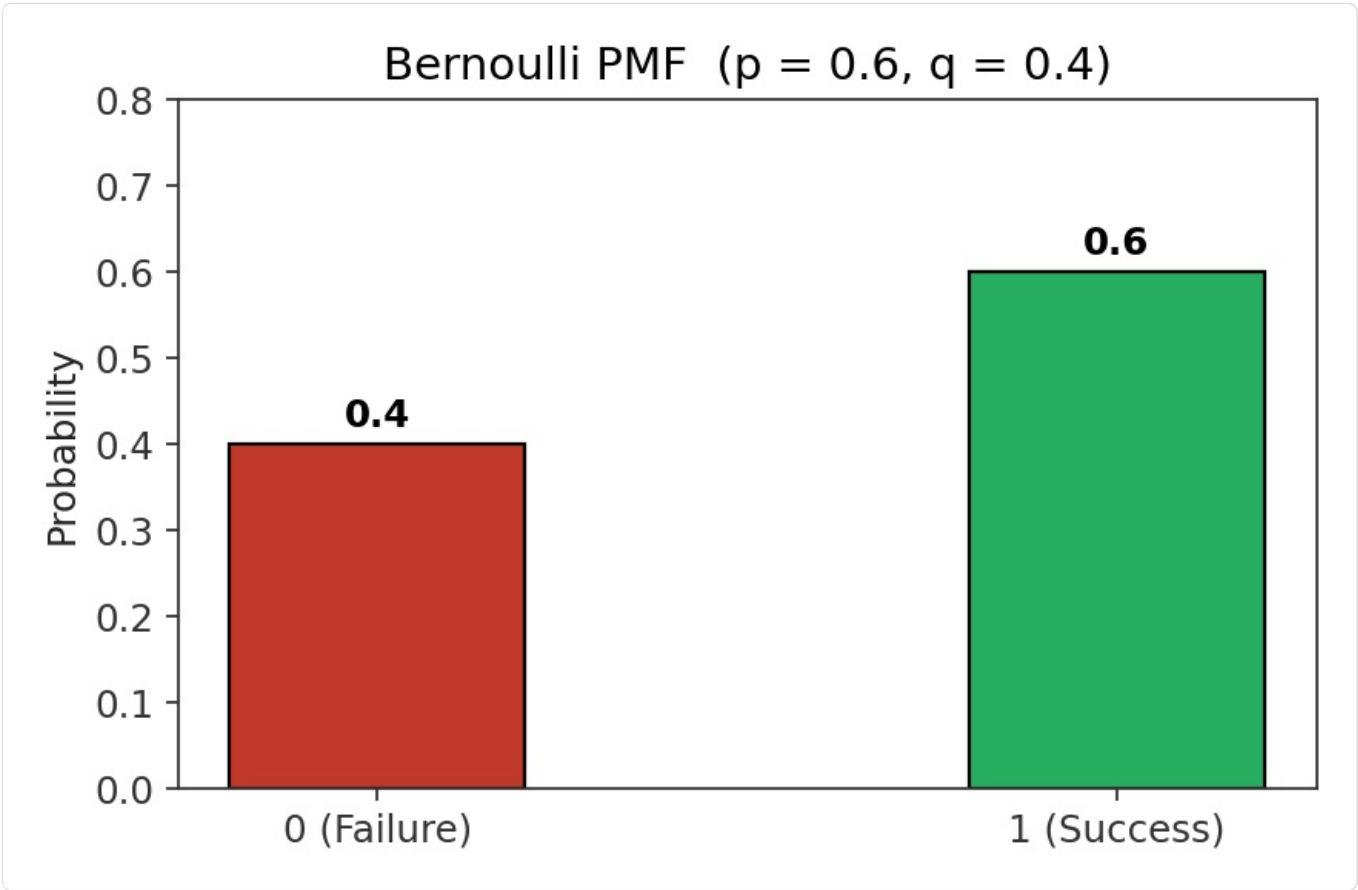


Fig 2.1 — Bernoulli PMF with $p = 0.6$ (60% chance of success).

Worked Example — App Usage

A company launches Smartphone "A". 60% of buyers use it (success), 40% don't (failure).

$\Pr(X = \text{Use}) = p = 0.6$ $\Pr(X = \text{Not Use}) = q = 1 - 0.6 = 0.4$

2.1 Mean, Median, Mode, Variance

Mean: $E(X) = \sum k \cdot p(k) = 0 \times q + 1 \times p = p$

Variance: $\sigma^2 = p \cdot q$ Std. Deviation: $\sigma = \sqrt{pq}$

Statistic	Rule
Median	0 if $p < 0.5$, [0,1] if $p = 0.5$, 1 if $p > 0.5$
Mode	p if $p > q$, otherwise q

Worked Example — Variance Calculation

Using $p = 0.6$, $q = 0.4$:

$\sigma^2 = 0.40 \times (0 - 0.6)^2 + 0.6 \times (1 - 0.6)^2 = 0.40 \times 0.36 + 0.6 \times 0.16 = 0.144 + 0.096 = \mathbf{0.24}$

(Same as $p \times q = 0.6 \times 0.4 = 0.24$ ✓)

✳ Extra Examples (Bernoulli in Real Life)

- **Email spam filter:** $\Pr(\text{spam}) = 0.15$, $\Pr(\text{not spam}) = 0.85$
- **Medical test result:** $\Pr(\text{positive}) = p$, $\Pr(\text{negative}) = 1-p$
- **ML classification:** A logistic regression model's output for a single sample is modeled as Bernoulli — this is exactly why "Binary Cross-Entropy Loss" is built on the Bernoulli likelihood!

3. Binomial Distribution

Definition: Models the number of successes in **n independent Bernoulli trials**, each with the same success probability **p**. When **n = 1**, the Binomial distribution reduces to the Bernoulli distribution.

Notation: $X \sim B(n, p)$

Parameter	Meaning
$n \in \{0,1,2,\dots\}$	Number of trials / experiments
$p \in [0,1]$	Success probability for each trial
$q = 1 - p$	Failure probability
$k \in \{0,1,2,\dots,n\}$	Number of successes (support)

PMF: $\Pr(k; n, p) = {}^nC_k \cdot p^k \cdot (1-p)^{n-k}$

where ${}^nC_k = n! / [k!(n-k)!]$ (Binomial Coefficient)

Mean = $n \cdot p$ Variance = $n \cdot p \cdot q$ Std. Dev. = $\sqrt{n \cdot p \cdot q}$

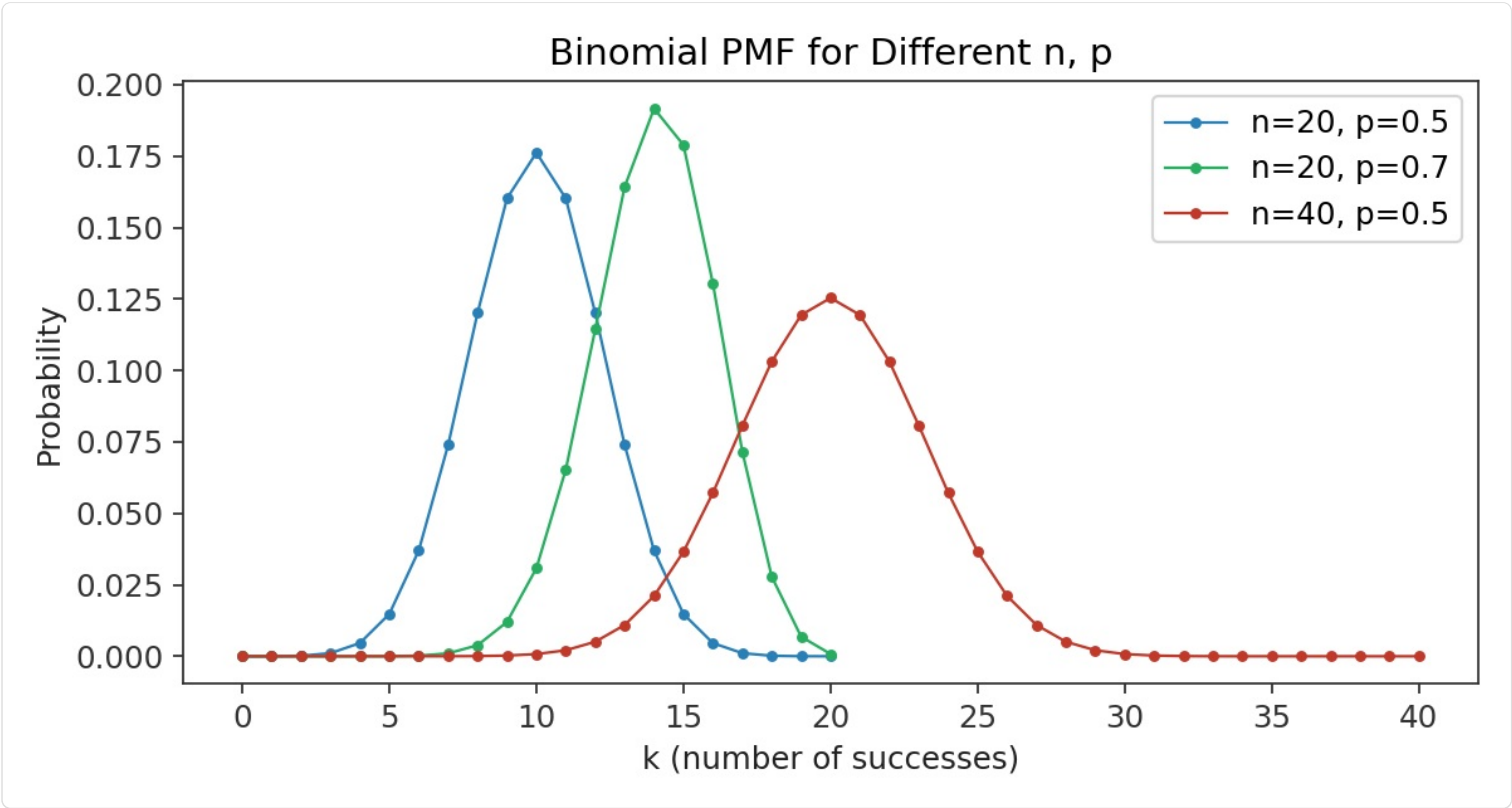


Fig 3.1 — Binomial PMF for various n and p. Notice it becomes more bell-shaped (like Normal) as n grows.

Worked Example 1 — Coin Flips

5 coin flips ($n=5$), $p = 0.5$. What is $\Pr(\text{exactly 3 heads})$?
 $\Pr(X=3) = {}^5C_3 (0.5)^3(0.5)^2 = 10 \times 0.125 \times 0.25 = \mathbf{0.3125}$ (31.25%)

Worked Example 2 — Quality Control

Inspecting 10 items ($n=10$), each item has 10% chance of being defective ($p=0.1$). What is $\Pr(\text{exactly 2 defective})$?
 $\Pr(X=2) = {}^{10}C_2 (0.1)^2(0.9)^8 \approx \mathbf{0.1937}$ (19.37%)

*** Extra Example — Exam MCQs**

A student randomly guesses on 8 MCQs ($n=8$), each with 4 options ($p = 0.25$ chance of guessing correctly).
Mean number of correct guesses = $n \cdot p = 8 \times 0.25 = \mathbf{2 \text{ questions}}$
Variance = $n \cdot p \cdot q = 8 \times 0.25 \times 0.75 = \mathbf{1.5}$
 $\Pr(\text{exactly 5 correct}) = {}^8C_5(0.25)^5(0.75)^3 \approx \mathbf{2.3\%}$ — showing that scoring well by pure guessing is very unlikely.

Bernoulli vs Binomial: Bernoulli = ONE trial. Binomial = MANY independent Bernoulli trials, counting total successes.

4. Poisson Distribution

Definition: Models the number of events occurring in a **fixed interval of time (or space)**, given a known constant average rate λ (**lambda**), where events happen independently of each other.

Property	Value
Type	Discrete Random Variable (PMF)
Describes	Number of events in a fixed interval
λ	Expected number of events per interval (rate)

PMF: $\Pr(X = x) = (e^{-\lambda} \lambda^x) / x!$

Mean = $E(X) = \mu = \lambda \times t$ (t = time interval) Variance = λ

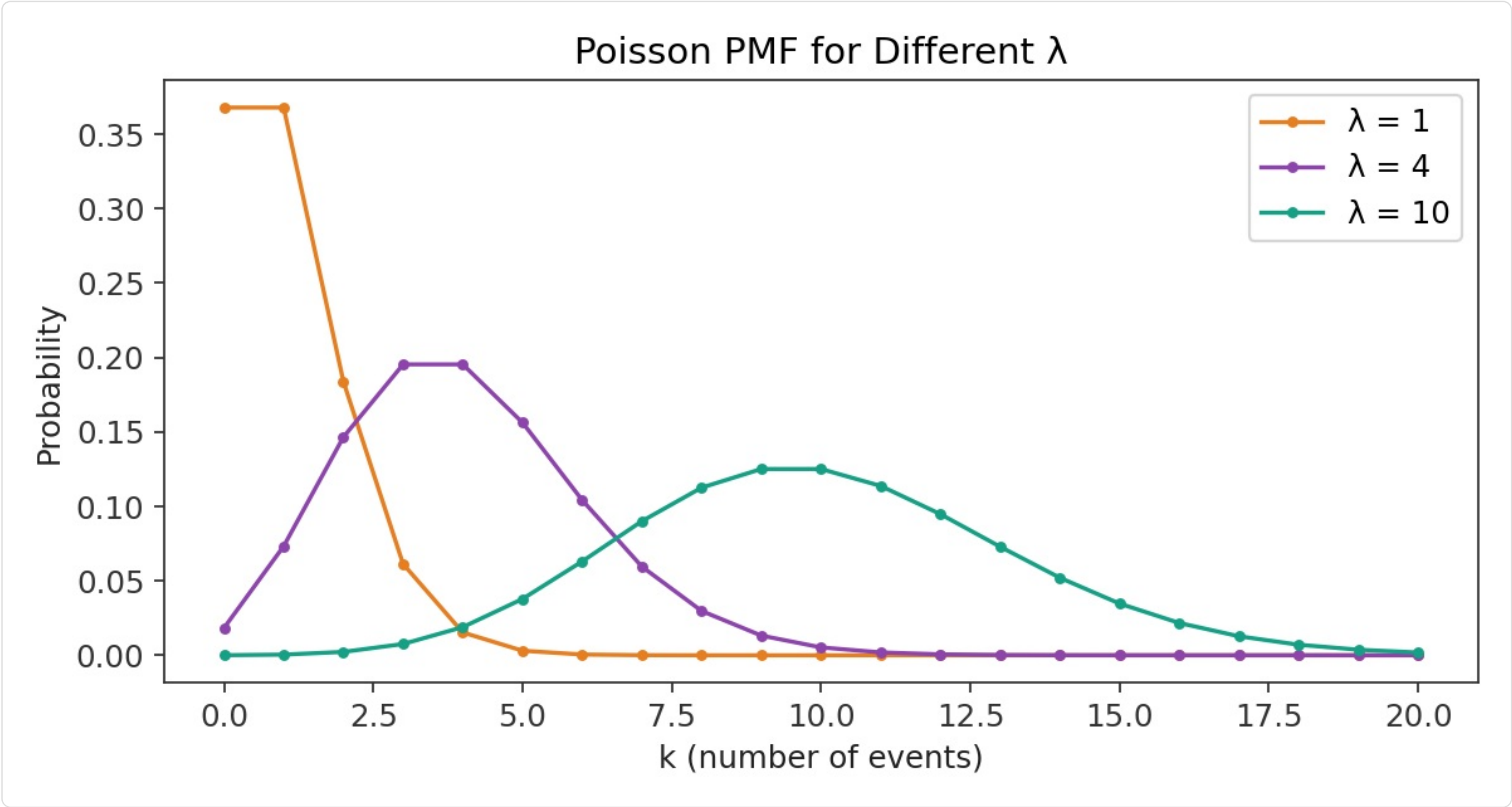


Fig 4.1 — Poisson PMF for $\lambda = 1, 4, 10$. As λ increases, the shape becomes more symmetric (approaching Normal).

Worked Example — Bank Customers

A bank receives $\lambda = 3$ customers per hour on average. What is $\Pr(\text{exactly 5 customers arrive in an hour})$?
 $\Pr(X=5) = (e^{-3} \times 3^5) / 5! = (0.0498 \times 243) / 120 \approx \mathbf{0.101}$ (10.1%)

✳ Extra Example — Website Traffic

A website gets an average of $\lambda = 12$ visits per minute.

- $\Pr(\text{exactly 0 visits in a minute}) = e^{-12} \approx 0.000006$ (extremely rare)
- $\Pr(\text{exactly 12 visits})$ is the single most likely outcome, since $\text{Mean} = \lambda = 12$
- If we look at a 5-minute window instead, new $\lambda = 12 \times 5 = 60$ (since λ scales with time interval t)

Common real-world Poisson use-cases: number of typos per page, number of earthquakes per year, number of calls to a call-center per hour, number of mutations in a DNA strand per generation.

🔗 **Binomial vs Poisson:** Binomial counts successes out of a *fixed number of trials*. Poisson counts events in a *fixed interval of time/space* with no upper limit on how many can occur. In fact, Poisson is the limiting case of Binomial when $n \rightarrow \infty$ and $p \rightarrow 0$, while $n \cdot p = \lambda$ stays constant.

5. Normal / Gaussian Distribution

Definition: A continuous probability distribution for a real-valued random variable that is symmetric about its mean — forming the famous "bell curve." Most natural and measured phenomena (height, weight, exam scores, measurement errors) follow this distribution.

Notation: $X \sim N(\mu, \sigma^2)$

Parameter	Meaning
$\mu \in \mathbb{R}$	Mean (center of the curve)
$\sigma^2 \in \mathbb{R} > 0$	Variance (spread of the curve)
$x \in \mathbb{R}$	Any real value

PDF: $f(x) = [1 / (\sigma\sqrt{2\pi})] \cdot e^{-\frac{1}{2} ((x-\mu)/\sigma)^2}$

Mean: $\mu = \Sigma x_i / n$ Variance: $\sigma^2 = \Sigma (x_i - \mu)^2 / n$ $\sigma = \sqrt{\text{Variance}}$

Key property: For a Normal distribution, Mean = Median = Mode, and the curve is perfectly symmetric — a "Bell Curve".

Normal / Gaussian PDF — Effect of μ and σ^2

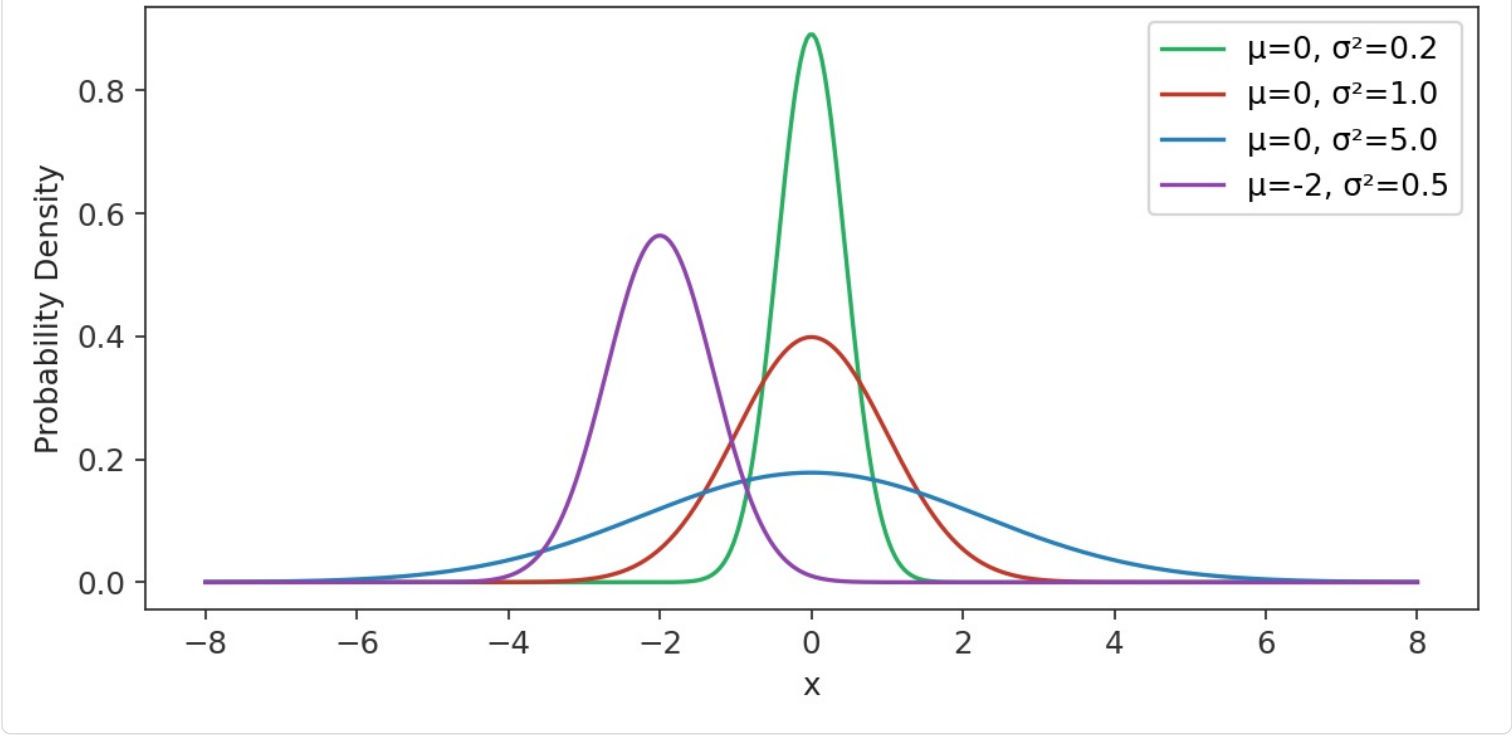


Fig 5.1 — As σ^2 increases, the curve gets wider (more spread out) and shorter.

Real-World Examples

- Weights of students in a class (medically healthy population)
- Heights of students in a class
- Iris dataset: Petal Length, Sepal Length, Petal Width, Sepal Width (used heavily by ML researchers as a teaching dataset)

5.1 The Empirical Rule (68-95-99.7 Rule)

Empirical Rule (68-95-99.7 Rule)

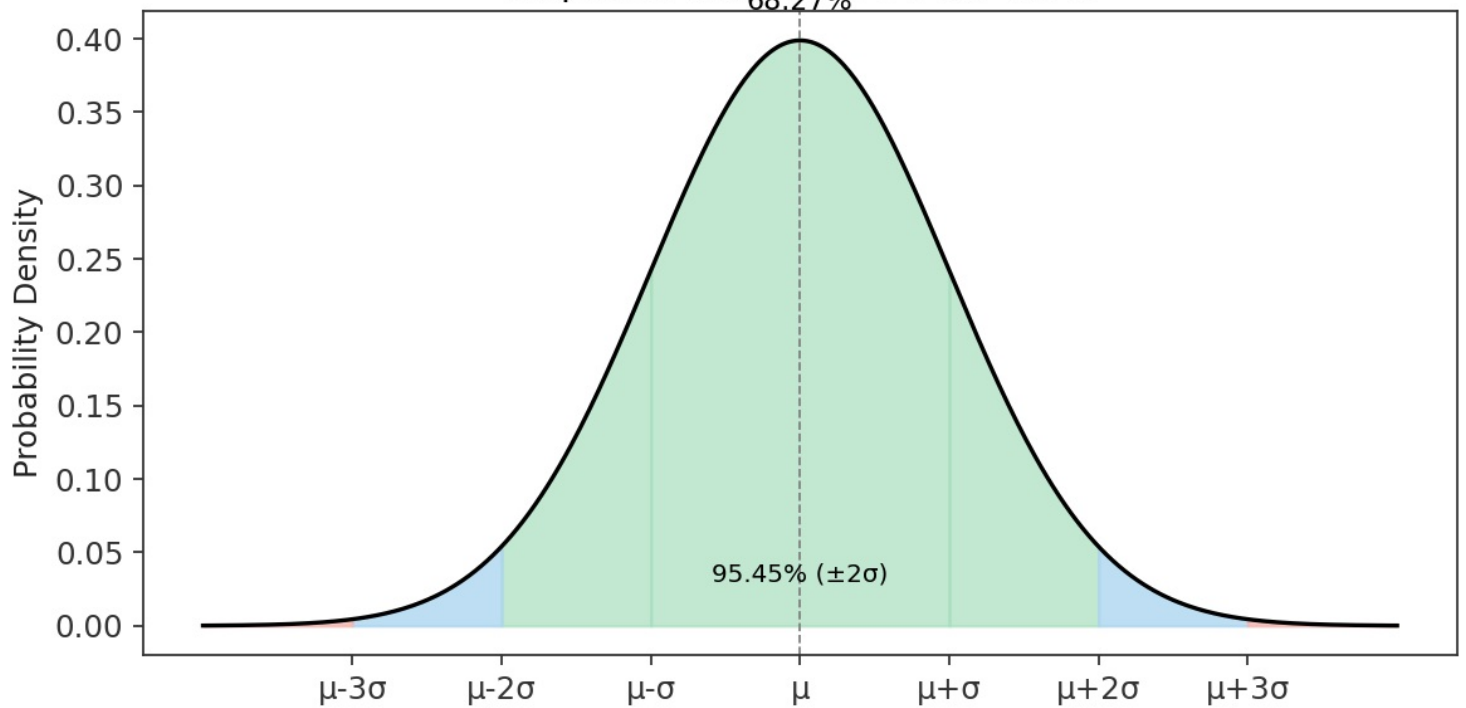


Fig 5.2 — The 68-95-99.7 rule tells us exactly what % of data lies within k standard deviations of the mean.

$$\begin{aligned} \Pr(\mu - \sigma \leq X \leq \mu + \sigma) &\approx 68\% \\ \Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) &\approx 95\% \\ \Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) &\approx 99.7\% \end{aligned}$$

* Extra Example — Exam Scores

Suppose exam scores are Normal with $\mu = 70$, $\sigma = 10$.

- 68% of students scored between 60 and 80 ($\mu \pm \sigma$)
- 95% of students scored between 50 and 90 ($\mu \pm 2\sigma$)
- A student who scored 100 is $(100 - 70)/10 = 3$ standard deviations above the mean — extremely rare (top ~0.15%)!

This rule is also how outlier detection often works in Data Science: anything beyond $\mu \pm 3\sigma$ is frequently flagged as an anomaly.

💡 **QQ Plot:** A Quantile-Quantile plot is used to visually check whether a dataset follows a Normal distribution — if the points fall roughly on a straight diagonal line, the data is approximately Normal.

6. Standard Normal Distribution & Z-score

Definition: The Standard Normal Distribution is a special Normal distribution with $\mu = 0$ and $\sigma = 1$, written $SND(\mu=0, \sigma=1)$ or $N(0,1)$. Any Normal distribution can be converted into it using the **Z-score** transformation.

$$\text{Z-score: } Z = (x_i - \mu) / \sigma$$

The Z-score tells you **how many standard deviations** a value is away from the mean. This process is also called **Standardization** — a critical preprocessing step in Machine Learning before feeding data to models like clustering, linear regression, or logistic regression.

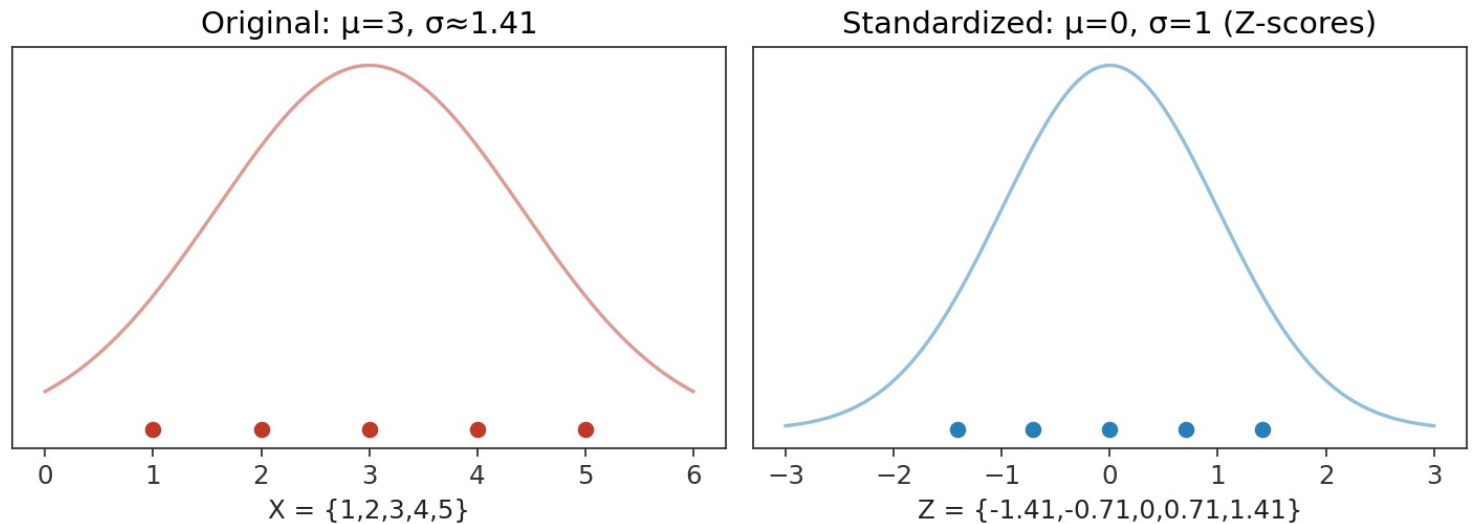


Fig 6.1 — Any Normal distribution (left) can be re-centered to $\mu=0, \sigma=1$ (right) using Z-scores.

Worked Example

$X = \{1, 2, 3, 4, 5\} \rightarrow \mu = 3, \sigma \approx 1.414$

Z-scores: $(1-3)/1.414 = -1.41$, $(2-3)/1.414 = -0.71$, $(3-3)/1.414 = 0$, $(4-3)/1.414 = 0.71$, $(5-3)/1.414 = 1.41$

So $Y = \{-1.41, -0.71, 0, 0.71, 1.41\}$ and now $X \sim SND(\mu=0, \sigma=1)$

Worked Example — "How many σ away?"

Given $\mu = 4, \sigma = 1$:

For $x = 4.25 \rightarrow Z = (4.25 - 4)/1 = \mathbf{0.25}$ (a quarter of a std. dev above the mean)

For $x = 2.5 \rightarrow Z = (2.5 - 4)/1 = \mathbf{-1.5}$ (one and a half std. dev below the mean)

6.1 Standardization for Machine Learning

Suppose we have a dataset with Age, Weight, Height, and Salary — all in very different scales (years, kg, cm, PKR/INR). Feeding raw values into distance-based ML models (like clustering or KNN) would let the largest-scale feature (Salary) dominate. So we standardize every column:

$$Z_{\text{age}} = (X_i - \mu_{\text{age}}) / \sigma_{\text{age}} \quad Z_{\text{weight}} = (X_i - \mu_{\text{weight}}) / \sigma_{\text{weight}}$$

After standardization, every feature has mean 0 and std. dev 1 — making them directly comparable. This feeds cleanly into: **(1) Clustering algorithms, (2) Linear Regression, (3) Logistic Regression.**

* Extra Example — Comparing Different Scales

A student scores 85/100 in Math (class $\mu=70, \sigma=10$) and 40/50 in Physics (class $\mu=32, \sigma=5$).

$Z_{\text{math}} = (85-70)/10 = 1.5$ $Z_{\text{physics}} = (40-32)/5 = 1.6$

Even though the raw Physics score (40) looks smaller than Math (85), the Physics Z-score is actually **higher** — meaning the student performed *relatively better* in Physics compared to their classmates!

7. Uniform Distribution

Two flavors: **Continuous Uniform** (PDF) for continuous variables, and **Discrete Uniform** (PMF) for discrete variables. In both, every outcome/interval is **equally likely**.

7.1 Continuous Uniform Distribution

An experiment where the outcome lies with equal likelihood anywhere between bounds a (minimum) and b (maximum).

Notation: $U(a, b)$ **Parameters:** $-\infty < a < b < \infty$

PDF: $f(x) = 1/(b-a)$ for $x \in [a, b]$; 0 otherwise

CDF: $F(x) = 0$ for $x < a$; $(x-a)/(b-a)$ for $x \in [a, b]$; 1 for $x > b$

Mean = $(a+b)/2$ Median = $(a+b)/2$ Variance = $(b-a)^2/12$

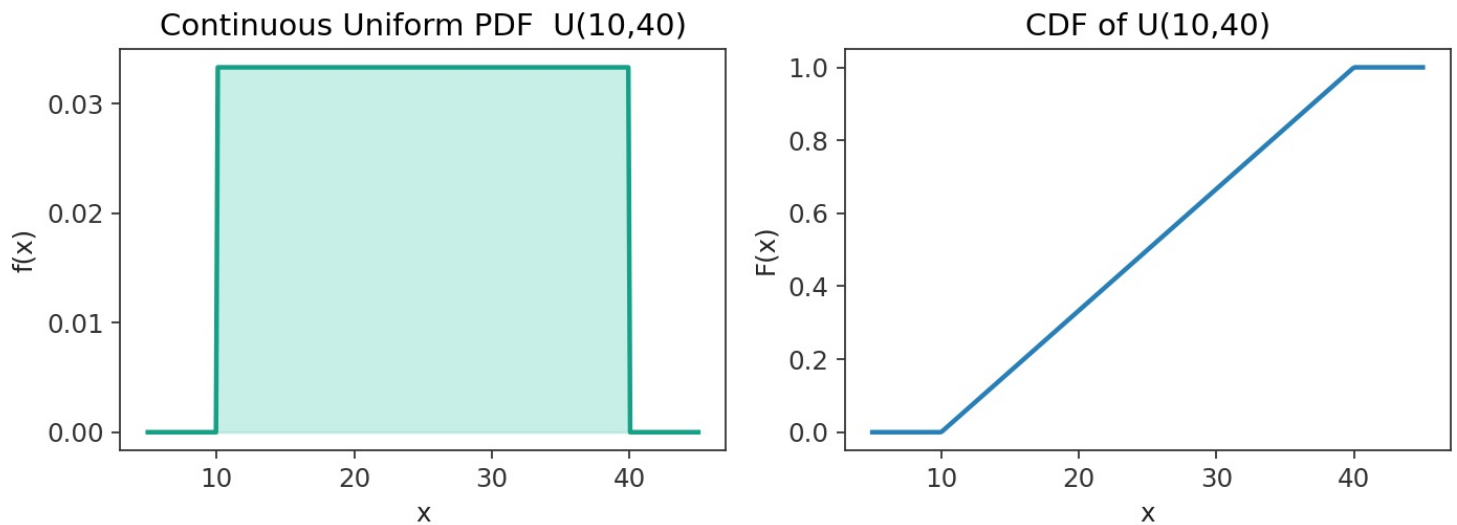


Fig 7.1 — Continuous Uniform PDF (flat rectangle) and its CDF (straight ramp).

Worked Example — Candy Sales

Daily candy sales are uniformly distributed between $a=10$ and $b=40$.

$\Pr(15 \leq X \leq 30) = (30-15) \times 1/(40-10) = 15 \times 1/30 = \mathbf{0.5}$ (50%)

$\Pr(X \geq 20) = (40-20) \times 1/30 = 20/30 \approx \mathbf{0.667}$ (66.7%)

7.2 Discrete Uniform Distribution

A finite number of values are all equally likely, each with probability $1/n$.

Notation: $U(a, b)$ **Parameters:** a, b where $b \geq a$, $n = b - a + 1$

PMF: $f(x) = 1/n$ Mean = $(a+b)/2$ Median = $(a+b)/2$

Discrete Uniform PMF — Fair Die U(1,6)

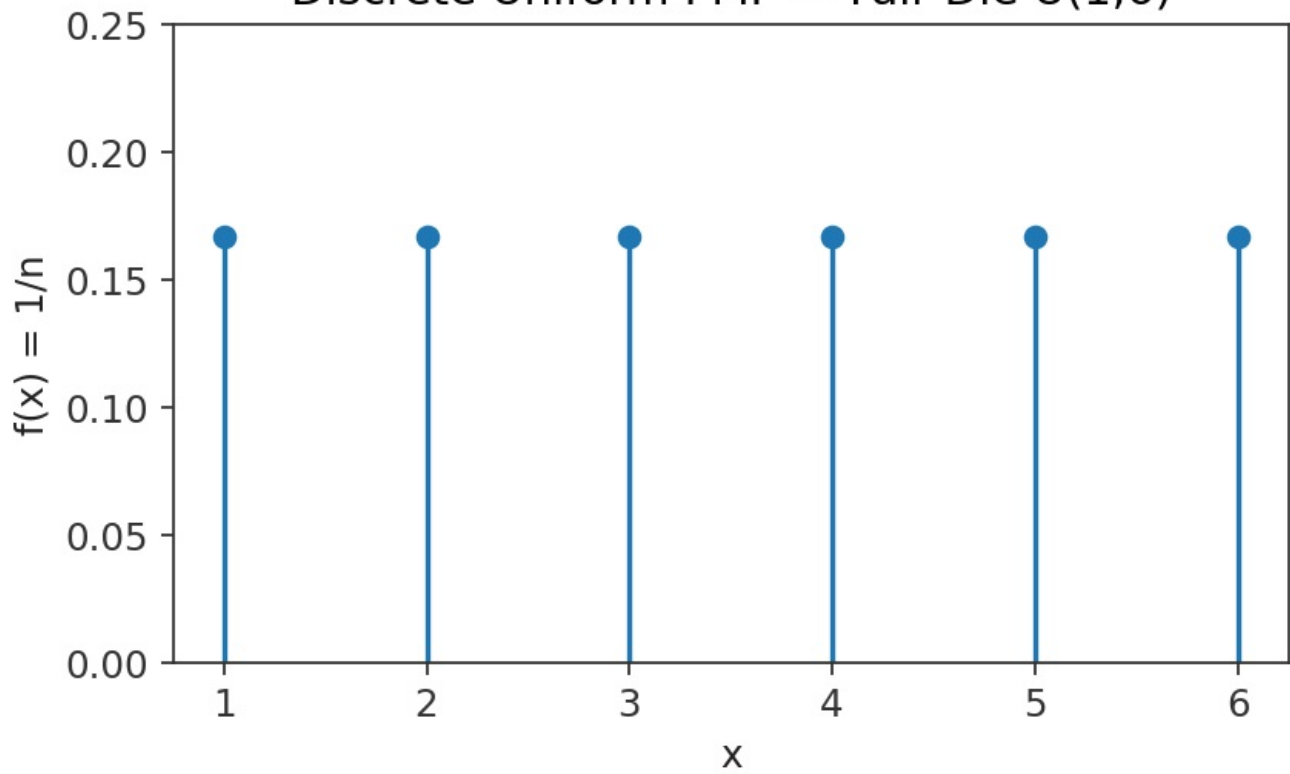


Fig 7.2 — Discrete Uniform PMF: rolling a fair die, each face has probability $1/6$.

Worked Example — Fair Die

Rolling a die: $a=1$, $b=6$, so $n = 6-1+1 = 6$, and each outcome has probability $1/6$.

* Extra Examples

- **Continuous:** The exact arrival time of a bus that comes "sometime between 2:00 and 2:20" — every second in that window is equally likely.
- **Discrete:** A random number generator picking an integer between 1 and 100 (each has probability $1/100$) — this is the backbone of shuffling algorithms and simple simulations.

8. Log-Normal Distribution

Definition: A continuous probability distribution of a random variable X whose **logarithm** is normally distributed. In other words: if $Y = \ln(X)$ is Normal, then X is Log-Normal. Equivalently, if Y is Normal, then $X = \exp(Y)$ is Log-Normal.

$$X \sim \text{LogNormal}(\mu, \sigma) \iff Y = \ln(X) \sim \text{Normal}(\mu, \sigma) \iff X = \exp(Y)$$

Shape: Right-skewed (long tail to the right) — unlike the symmetric Normal curve.

Log-Normal PDF — Right-Skewed Distribution

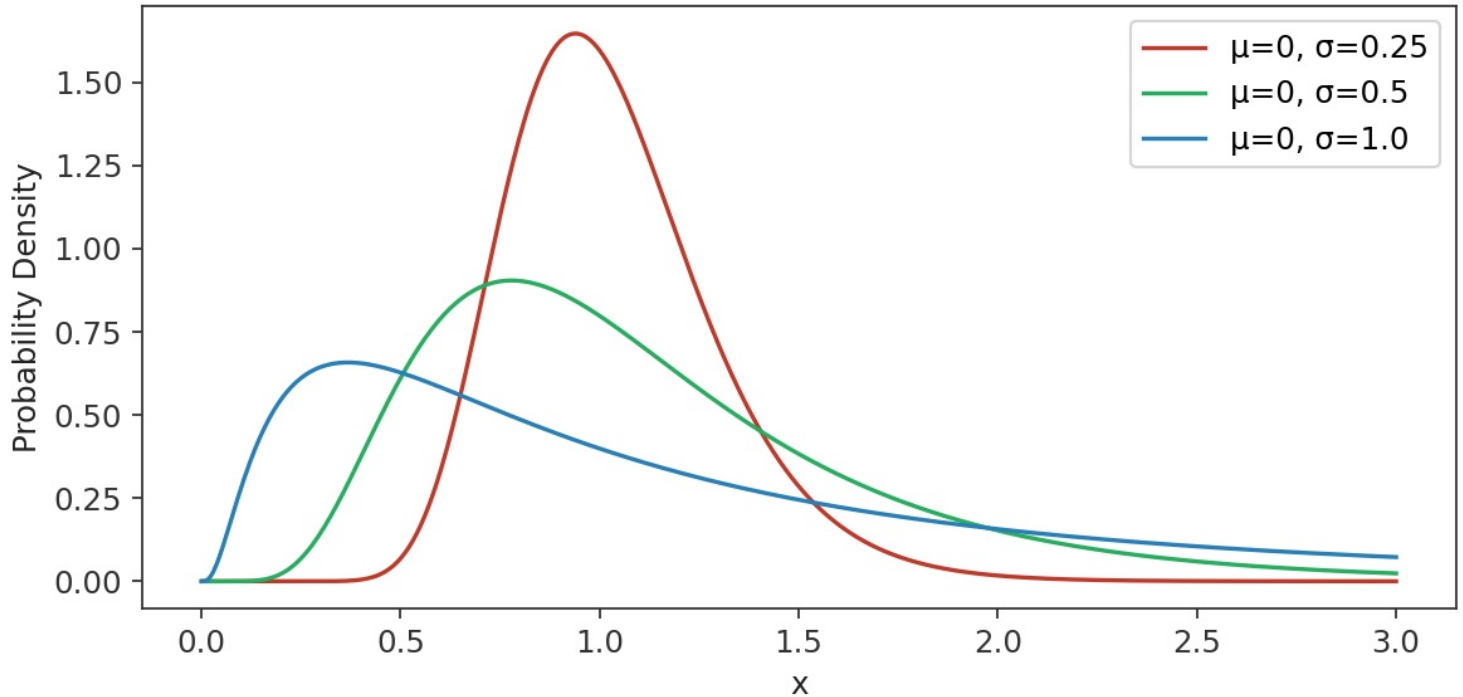
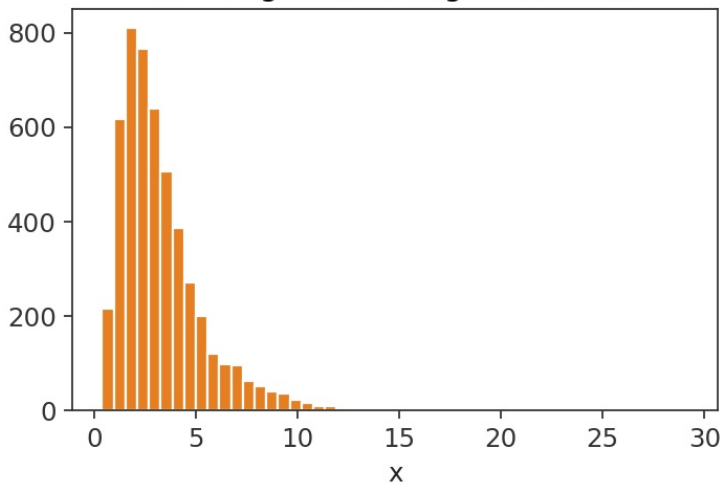


Fig 8.1 — Log-Normal PDF for different σ (same $\mu=0$). Always right-skewed, never negative.

$X \sim \text{Log-Normal}$ (right-skewed)



$Y = \ln(X) \sim \text{Normal}$

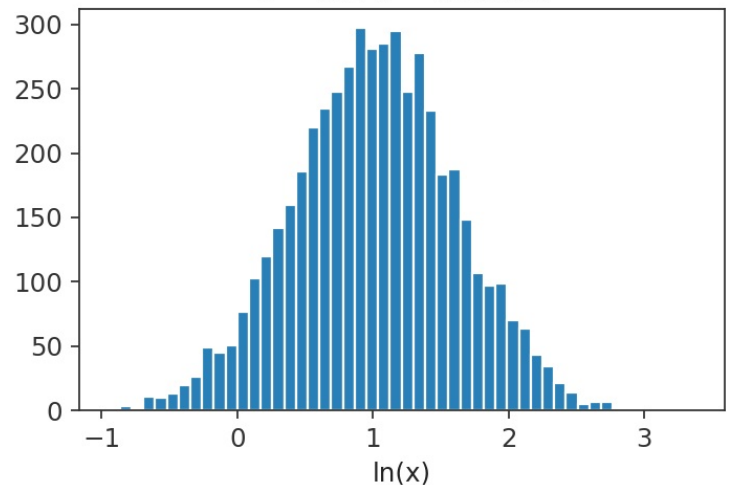


Fig 8.2 — Taking the natural log of Log-Normal data (left) produces a Normal distribution (right). This is exactly how a QQ-plot check works after log-transforming skewed data.

Real-World Examples

- **Wealth distribution of the world** — Bill Gates, Adani, Elon Musk sit far in the long right tail
- **Discussion forums** — length of comments
- **Length of chess games**
- **Dwell time on online articles/jokes/news**
- **Salaries of employees** in a company (most cluster around a typical range, a few executives earn far more)

* Extra Example — City Populations

The population sizes of cities across a country typically follow a Log-Normal distribution: many small towns, fewer medium cities, and very few mega-cities (like Karachi or Lahore). If you take $\ln(\text{population})$ for every city, the resulting histogram looks approximately Normal — this trick (log-transformation) is used constantly in Data Science to "fix" skewed features before applying models that assume normality (like Linear Regression).

💡 Why does this matter in Data Science? Many real datasets (income, prices, view-counts) are Log-Normal, not Normal. Applying `np.log()` to such a column before modeling is one of the most common feature-engineering tricks.

9. Power Law Distribution

Definition: A functional relationship where a relative change in one quantity produces a proportional relative change in another — **one quantity varies as a power of another**, independent of the initial size of the quantities.

80-20 Rule (Pareto Principle)

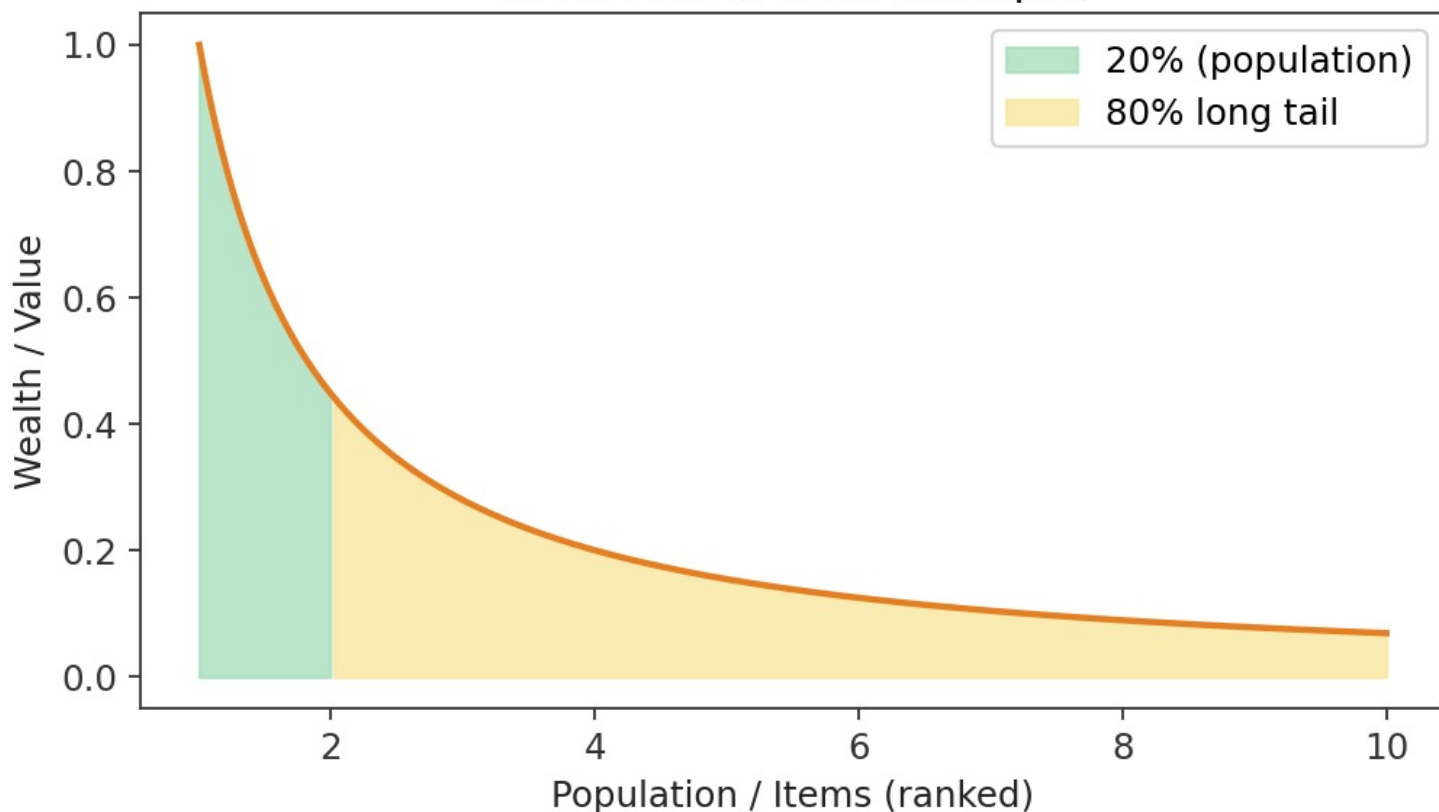


Fig 9.1 — The classic 80-20 shape: a small fraction of items account for the majority of the value (the "long tail" trails off to the right).

The 80-20 Rule in Action (IPL Example)

- 20% of teams are responsible for winning 80% of matches
- 80% of wealth is distributed within 20% of the total population
- 80% of the total oil reserves belong to 20% of nations
- Word frequencies in most languages follow a power law (a few words are used extremely often)
- 20% of major software defects cause 80% of upcoming defects in a slow product

Transformation trick: Power Law data → apply $\ln(x)$ or a **Box-Cox Transformation** → becomes approximately Normal. This is then checked using a **QQ Plot**.

* Extra Example — YouTube Views

On YouTube, a tiny fraction of videos (viral content) get the overwhelming majority of total views, while millions of videos get almost none. This "few dominate, many are negligible" pattern is a textbook Power Law — and it's also why simple averages are misleading for such data (the mean is dragged up by a handful of outliers; the median is far more representative).

10. Pareto Distribution

Definition: Named after Italian economist **Vilfredo Pareto**, this is a specific **power-law probability distribution** historically used to describe the distribution of wealth in a society — a large portion of wealth is held by a small fraction of the population. It is classified as a **Non-Gaussian** distribution.

Pareto Type I PDF has shape parameter α (alpha).
As α increases $\uparrow$, the curve gets steeper near the minimum value.

Pareto (Power-Law) Distribution — Type I

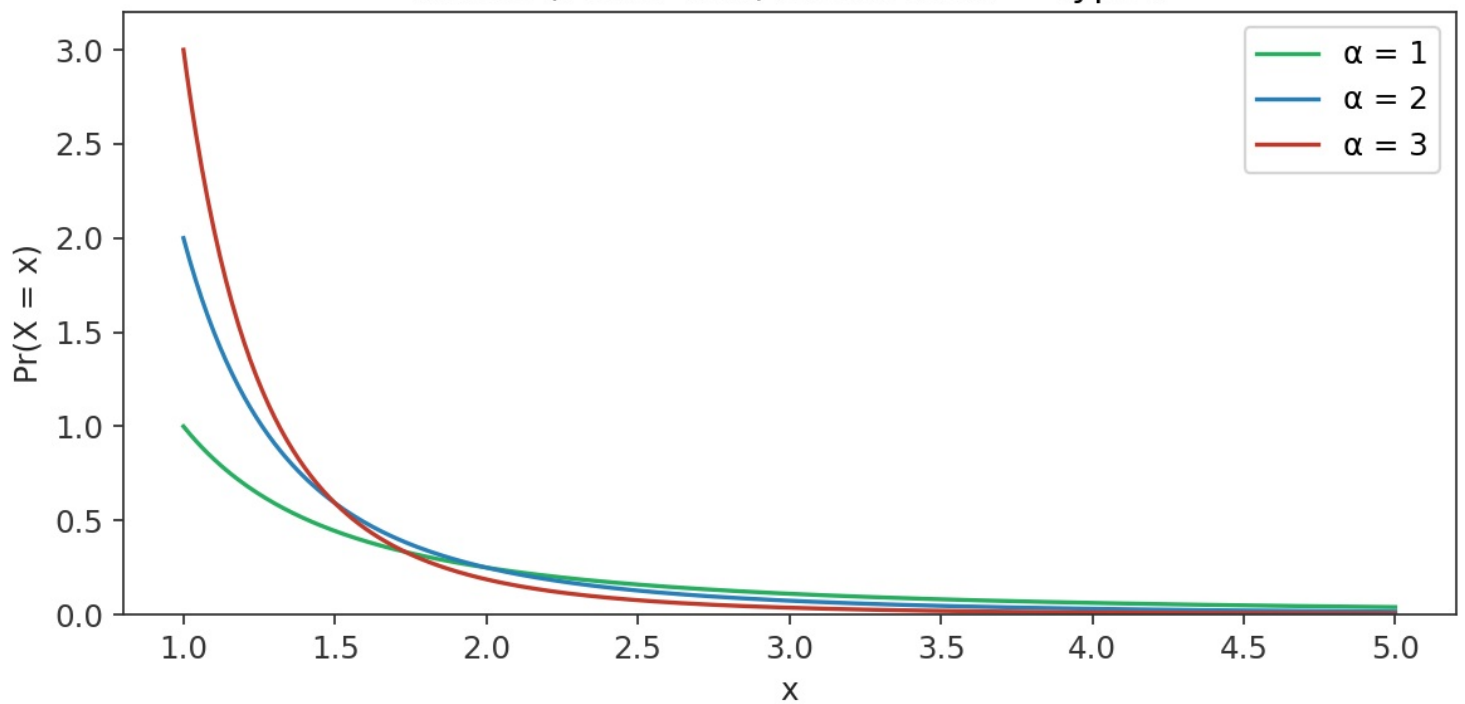


Fig 10.1 — Pareto Type I PDF for $\alpha = 1, 2, 3$. Larger α means probability concentrates faster near the minimum.

Transformation pipeline: Pareto/Power-Law (right-skewed, long tail) $\rightarrow$ Log-Normal transform / Box-Cox Transformation $\rightarrow$ Approximately Normal $\rightarrow$ Verify with a QQ Plot.

Real-World Example — IT Industry

- 80% of the entire project is completed by 20% of the team (your top performers)
- 80% of defects can be solved by fixing 20% of the root-cause bugs

* Extra Example — Customer Revenue

An e-commerce company finds that 20% of its customers ("VIP" buyers) generate 80% of total revenue. This is a Pareto pattern. Businesses use this insight to prioritize retention marketing toward that top 20% rather than spreading the budget evenly across all customers — since the marginal value of retaining a VIP customer is far higher.

🔗 **Power Law vs Pareto:** "Power Law" is the general mathematical family of distributions ($y \propto x^{-k}$). "Pareto" is a specific, well-studied member of that family, most associated with the 80-20 rule.

11. Central Limit Theorem (CLT)

Definition: The CLT relies on the concept of a **sampling distribution** — the probability distribution of a statistic (like the mean) computed over many samples drawn from a population.

The Big Idea: The sampling distribution of the **mean** will always be approximately **Normal**, as long as the sample size is large enough (commonly $n > 30$) — **REGARDLESS** of whether the original population is Normal, Poisson, Binomial, Skewed, or any other shape!

If $X \sim N(\mu, \sigma) \rightarrow$ Sampling distribution of mean: $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$

If X does not follow $N(\mu, \sigma)$ but $n > 30 \rightarrow$ CLT still applies $\rightarrow \bar{x} \sim N(\mu, \sigma/\sqrt{n})$

where σ = population standard deviation, μ = population mean, n = sample size.

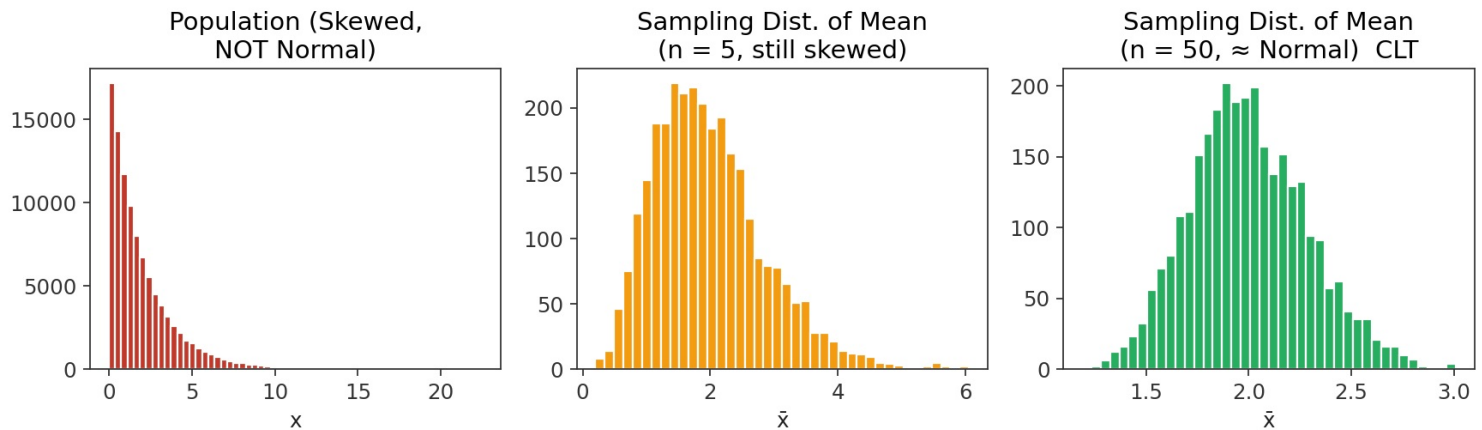


Fig 11.1 — Left: a heavily skewed (non-Normal) population. Middle: sampling distribution of the mean with small samples ($n=5$) — still somewhat skewed. Right: with larger samples ($n=50$) the sampling distribution of the mean becomes clearly Normal-shaped — this IS the Central Limit Theorem in action.

11.1 How it works, step by step

1. Take many random samples $S_1, S_2, S_3, \dots, S_m$ from the population, each of size n .
2. Compute the mean of each sample: $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_m$
3. Plot these sample means — this new distribution is the "**sampling distribution of the mean**"
4. As n grows large ($n > 30$ is the common rule of thumb), this sampling distribution becomes Normal.

Worked Example

A factory's bulb lifespans follow a heavily right-skewed distribution (some bulbs fail fast, most last a normal time, a few last extremely long) with population mean $\mu = 1000$ hours and $\sigma = 200$ hours.

If we take samples of $n = 100$ bulbs repeatedly and compute each sample's average lifespan, those *averages* will be approximately Normal: $\bar{x} \sim N(1000, 200/\sqrt{100}) = N(1000, 20)$.

* Extra Example — Why CLT is a Big Deal in Data Science

CLT is the mathematical foundation behind:

- **Confidence intervals** and **hypothesis testing** (t-tests, z-tests) — these all assume the sampling distribution of the mean is Normal, which CLT guarantees even when raw data isn't.
- **A/B testing** in product analytics — comparing average conversion rates between two groups relies directly on CLT.
- **Bootstrapping** in machine learning validation techniques.

Without CLT, statisticians would need to know the exact population distribution before doing any inference — CLT frees us from that requirement as long as the sample is large enough.

12. Estimates — Point & Interval

Estimate: A specified, observed numerical value used to estimate an unknown population parameter (like the true population mean μ , which we can rarely measure directly).

12.1 Point Estimate

A **single numerical value** used to estimate an unknown population parameter.

Example
The sample mean ($\bar{x}$) is a point estimate of the true population mean (μ). E.g., you survey 100 people and find their average height is $\bar{x} = 65$ inches — this single number 65 is your point estimate of the true (unknown) population mean.

12.2 Interval Estimate

A **range of values** used to estimate the unknown population parameter — much more informative than a single point, because it also communicates uncertainty.

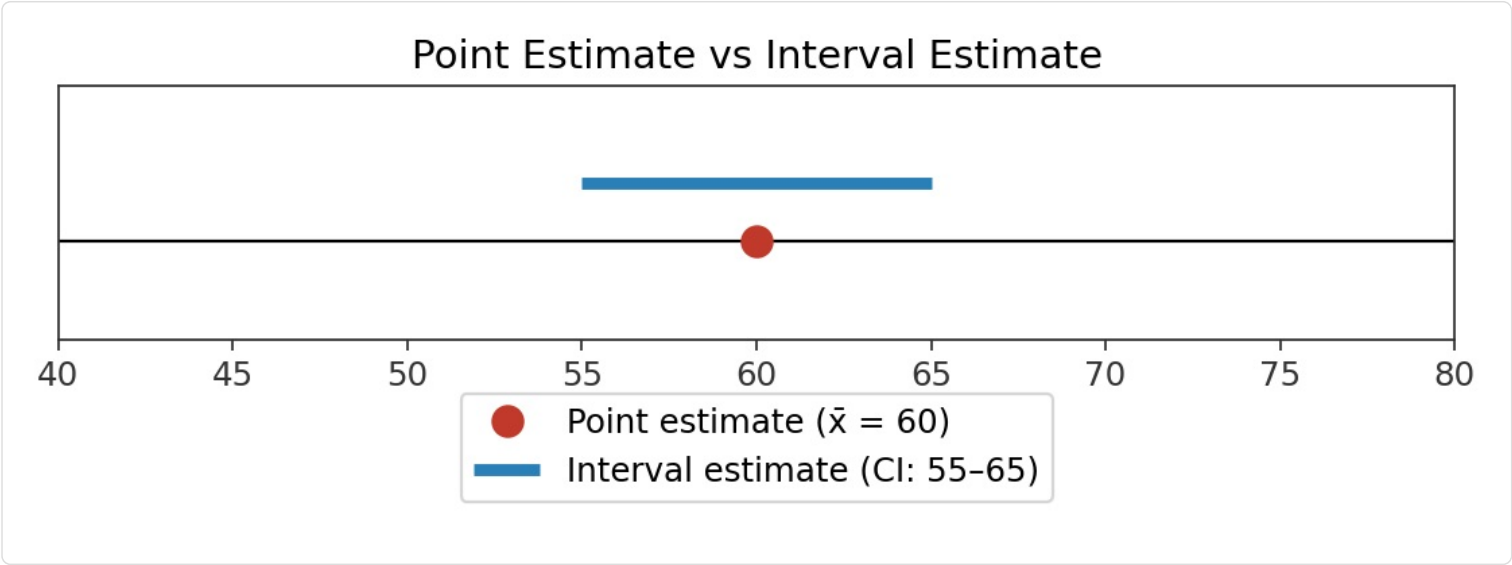


Fig 12.1 — A point estimate is one number; an interval (confidence) estimate is a range around it.

Example
Instead of just saying "the average height is 65 inches," we say: "We are 95% confident the true average height lies between **55 and 65**." This range [55, 65] is the **Confidence Interval**, and 65 sits as the point estimate inside it.

Aspect	Point Estimate	Interval Estimate
Form	Single number	Range (lower, upper bound)
Communicates uncertainty?	No	Yes
Example	$\bar{x} = 65$	[55, 65] at 95% confidence

✳ Extra Example — Election Polling

A news channel reports: "Candidate A is leading with **52% support**." That's a point estimate. A more rigorous report says: "Candidate A has 52% support, with a margin of error of $\pm 3\%$, i.e., a 95% confidence interval of [49%, 55%]." Because the interval dips below 50%, a statistician would caution that the race might actually be closer than the headline "52%" suggests — this is exactly why interval estimates matter more than point estimates alone.

💡 Interval estimates are built directly using the Central Limit Theorem — since we know the sampling distribution of the mean is Normal (for large n), we can compute exactly how wide the confidence interval should be around our point estimate.

13. Master Summary Table

Distribution	Type	Mean	Variance	Real-World Example
Bernoulli	Discrete (PMF)	p	pq	Coin flip, Pass/Fail
Binomial	Discrete (PMF)	np	npq	Number of defective items in a batch
Poisson	Discrete (PMF)	λt	λ	Customers arriving per hour
Discrete Uniform	Discrete (PMF)	$(a+b)/2$	—	Rolling a fair die
Normal / Gaussian	Continuous (PDF)	μ	σ^2	Height, weight, exam scores
Continuous Uniform	Continuous (PDF)	$(a+b)/2$	$(b-a)^2/12$	Bus arrival time in a window
Log-Normal	Continuous (PDF)	$\exp(\mu+\sigma^2/2)$	—	Income, stock prices, salaries
Power Law / Pareto	Continuous, non-Gaussian	—	—	Wealth distribution, viral content views

 **Quick Recall Checklist**

- PMF → Discrete | PDF → Continuous | CDF → Cumulative probability (both)
- Bernoulli = 1 trial, 2 outcomes → Binomial = n trials, counts successes
- Poisson = counts events in a fixed time/space interval given rate λ
- Normal = symmetric bell curve; 68-95-99.7 empirical rule; Z-score standardizes it
- Uniform = every outcome equally likely (flat PDF/PMF)
- Log-Normal = right-skewed; $\ln(X)$ turns it into a Normal distribution
- Power Law / Pareto = 80-20 rule; long right tail; log/Box-Cox transform → Normal
- CLT = sampling distribution of the mean is ALWAYS Normal for $n > 30$, regardless of population shape
- Point Estimate = single value | Interval Estimate = range with confidence level

Compiled & expanded study notes — includes original handwritten content plus additional worked examples for deeper understanding.